Rule of Sum:

If A_1 is an event that can be performed in m_1 ways, and event A_2 can be performed in m_2 ways. Further the events A_1, A_2 cannot happen at the same time. Then either of the two events (A_1 or A_2) can happen in (m_1+m_2) ways.

Rule of Product.

If A_i is an Event that can be performed in 'm' ways & A_2 can be performed in n ways then both A_1 & A_2 can be performed in mn ways [A_1 & A_2 be two events which happens one after another]

Note:- The above two can be extended for any finite number of tasks.

Rule of product

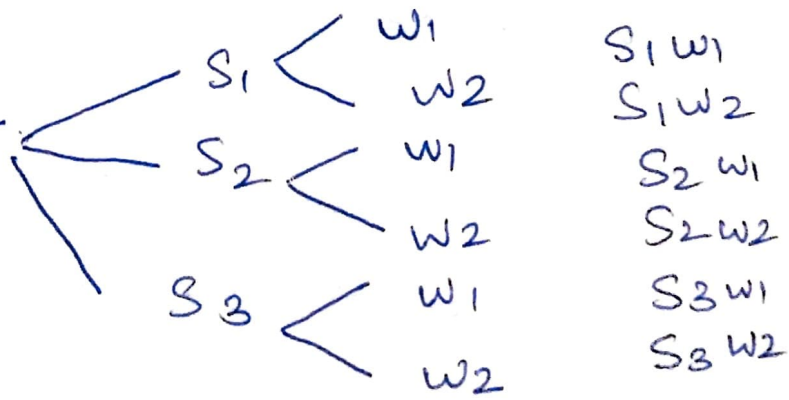
A student has 3 school Bags & 2 water bottles. Find the total no. of ways to carry schod bag and water bottle.

School Bags: S_1 S_2 S_3 — 3 ways

water bottles: W_1 W_2 — 2 ways

Total ways: $3 \times 2 = 6$ ways

(ii) Student



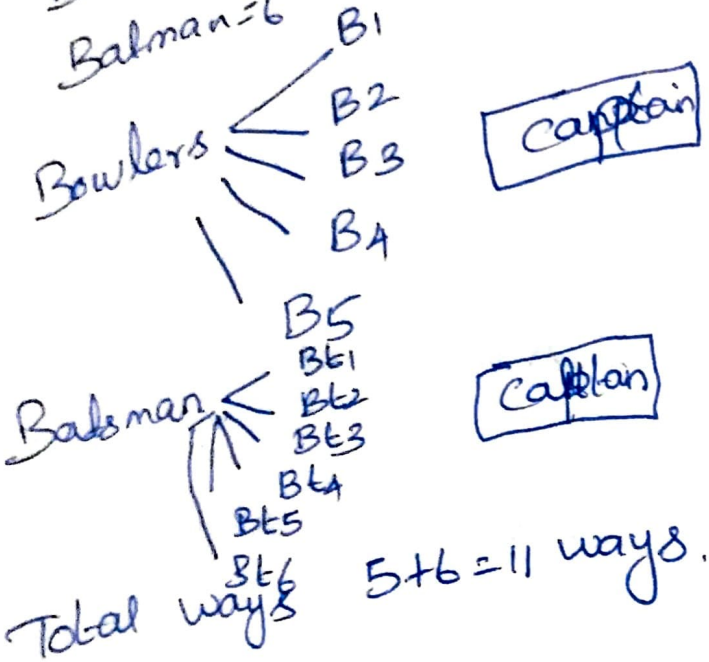
Rule of sum

Select Captain of Cricket team from 5 bowlers & 6 balsman.

Bowlers = 5

Balman = 6

Captain 5 / 6 = 11



Total ways 5+6 = 11 ways.

Examples :

1 Consider a problem of electing a representative from 30 boys & 20 girls.

(i) In how many ways a representative can be selected

(ii) In how many ways two representatives one from boys and another from girls can be selected.

No of ways to select a representative from boys = 30 = m.

No " " " " " from girls = 20 = n

∴ By the rule of sum, no of ways to select a representative = m+n = 30+20 = 50

By the rule of product no of ways to select two representatives (one from Boys & one from Girls)

(iv) $30 * 20 = 600$ ways.

② Manufacturing of licence pletes consisting of two digits followed by the four digits.

(i) If no letter or digit is allowed to repeat, then how many plates are possible?

(ii) with repetition of only letters how many plates are possible.?

(iii) with repeatition of letters and digits how many licence plates are possible.

(iv) If repeatitions are allowed, how many plates have only vowels & even digits)

Ans (i) The no of possible plates with no letter (or) digit is allowed to repeat is

$26 \times 25 \times 10 \times 9 \times 8 \times 7$.

(ii) Total no of licence plates with repeatition of letters is $= 26 \times 26 \times 10 \times 9 \times 8 \times 7$.

(iii) Total no of licence plates with repeatition of letters and digits is $= 26 \times 26 \times 10 \times 10 \times 10 \times$

(Iv) The no of possible licence plates with repeatitions of only vowels and even digits is $= 5 \times 5 \times 5 \times 5 \times 5 \times 5$

a e i o u → (5)

0 2 4 6 8 → (5)

③ Consider an event of distinct dice. Determine the no of ways of happening of either of the following two events:

Event 1: Getting even no's on both the dice

Event 2: sum of the numbers on the two dice is

Sol: Event 1: (2,2) (2,4) (2,6) (4,2) (4,4) (4,6) (6,2) (6,4) (6,6) (9 ways)

Event 2: (1,2) (1,4) (1,6) (2,1) (2,3) (2,5); (3,2) (3,4,) (3,6) (4,1) (4,3) (4,5) (5,2) (5,4) (5,6) (6,1) (6,3) (6,5) [18 ways]

∴ Total no of ways of happening Event 1 or Event 2 is 9+18=27 ways.

④ A hotel has 10 standard rooms and 6 deluxe rooms with complimentary breakfast of 2 types: South Indian with 4 items and continental with 3 items. A customer chose to have a standard room with South Indian items or deluxe room with continental items. Find the no of possible ways of either of the two choices.

Soln: The no of way of the 1st choice = 10×4 = 40

The no " " " 2nd choice = 6×3 = 18

The total no of ways = 40+18 = 58//

Permutations and combinations.

Permutations means arrangement of things where the word arrangement is to be understood in the perspective that the order of things is to be necessarily considered.

Combinations means selection of things where it is evident that the order is not in consideration

Defn Permutation [linear]

The linear arrangement of distinct objects of any linear is called permutation.

Given a collection of 'n' distinct objects of any linear arrangement of 'r' objects can be done in nP_r (or) $P(n,r)$ ways

(ie) $nP_r = P(n,r) = \frac{n!}{(n-r)!}$

Note:

If repetitions are allowed then there are n^r possible arrangements.

Ex: Find the no of words of three distinct letters formed from the letters of the word "JNTU". If repetitions are allowed, form the no of possible six letter sequence.

Soln. The no of words of three distinct letters form from the letters of the word "JNTU" is $4P_3 = \frac{4!}{(4-3)!} = \frac{4\times3\times2}{1!} = 24$.

If repetitions are allowed, the total no. of possible six letter sequence is $= 4^6$.

$_ _ _ _ _ _$

(2) In how many ways can eight men & eight women be seated in a row if
(i) any person may sit next to any other
(ii) Men and women must occupy alternative seats. (iii) Generalize this result for n men & n women

(i) The no of arrangements of any person may sit to any other $= {}^{16}P_{16} = \frac{16!}{(16-16)!} = 16!$

(ii) Men sitting first & this can be done in $8! \times 8!$ ways women sitting first can be done in $8! \times 8!$
Thus the no of ways men and women occupy alternative seats $= [8! \times 8!] + [8! \times 8!] = 2(8!)^2$.

(iii) By (ii) using (i) we've $2(n!)^2$ //.

$\because n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1$

(3) There are four different sizes of formal shirts, three different sizes of pants and two different sizes of 'blazers' to be arranged in the display show case of a textile show room. How many different arrangements are possible if
(i) all of them are put in the show case.
(ii) different category of wears are all together.
(iii) only formal shirts are together.

(i) all of them are simply to put together is possible in ${}^4P_4 \cdot {}^3P_3 \cdot {}^2P_2$.

4 shirts: 3 pants 2 blazers

(ii) $4! \times 3! \times 2! = 24 \times 6 \times 2 = 288$ ways.

④ If ~~repetition~~ the no of ~~possible 12-letter~~ ~~set~~ How many different arrangements of the letter is the word "BOUGHT" can be formed, if the vowels must be kept next to each other.

Ans The given word has vowels O, U. Now treat OU as a single letter:

The total no of letters = 6-2+1=5.

5 letter can be arranged in 5! ways. ch of these arrangements the vowels O, U n be arranged among themselves in 2! ways.

By the rule of product, the required no of arrangements = 5! x 2!

Permutations with Identical objects.

If there are 'n' objects of which n_1 are type 1. n_2 are type 2. - - - n_r are type r with $n_1+n_2+\cdots+n_r=n$. Then there are $\frac{n!}{n_1!\,n_2!\cdots n_r!}$ linear arrangements are possible.

Example:

① Find the no of permutations of letters of the word 'COLLEGE', RECURRENCE

C→1 O→1 L→2 E→2 G→1.

No of permutations $= \frac{7!}{2!\times 2!}$

~~2. Find the~~ RECURRENCE

R→3 E→3 C→2 U→1 N→1

$\frac{10!}{3!\times 3!\times 2!}$

② How many arrangements are there for the letters of word "SOCIOLOGICAL"

(i) In how many arrangements 'A' and 'G' are together (ii) In how many arrangements vowels are adjacents (iii) In how many " " " and sorted

S → 1 O → 3 C → 2 I → 2 L → 2 G → 1

There are 12 letters in the given word

Total no of arrangements $= \frac{12!}{3!\,2!\,2!\,2!}$

(i) If A and G are together then treat A and G as a single letter.

Total no of letters of A and G are adjacent.

$= 12 - 2 + 1 = 11$.

No of arrangements in which A and G are adjacent $= \frac{11!}{3! \times 2! \times 2! \times 2!} \times 2!$

[$\because 2!$ = No of arrange A & G (AG & GA)]

(ii) The vowels in the given word are A, I, O which are repeated 1, 2, 3 times respectively Treating all these vowels A, I, O in the given word as a single letter.

③ Find the no of letters of the word 'INSTITUTION'
How many of these (i) begin with I (ii) begin with I and end with N
How many arrangements (iii) three T's together (iv) three T's are not together.

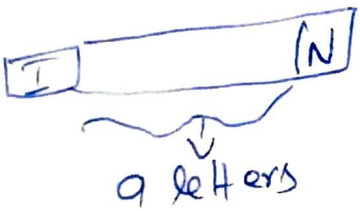
In Given word there are 11 letters.

I → 3, N → 2, T → 3, S → 1, U → 1, O → 1

∴ Total no of arrangements = $\frac{11!}{3! \times 3! \times 2!}$

(i) 

No of arrangements begin with I = $\frac{10!}{2!2!3!}$

(ii) 

No of arrangements begin with I & end with N = $\frac{9!}{2! \times 3!}$

(iii) Considers 3 T's are together as single arrange

then total no of letters = 11 − 3 + 1 = 9.

∴ Total no of arrangements in which 3 T's are together = $\frac{9!}{3! \times 2!}$

(iv) The no of arrangements in which 3 T's are not together = Total no of arrangements − No of arra in which 3 T's not togeth

$= \frac{11!}{3! \times 3! \times 2!} - \frac{9!}{3! \times 2!}$

④ How many +ve integers 'n' can we form using the digits 3, 4, 4, 5, 5, 6, 7, if we want 'n' to exceed 5,000,000?

Ans Let $n = x_1 x_2 x_3 x_4 x_5 x_6 x_7$

By data, 7 digits are given, Here 4 & 5 repeated twice and rest 3, 6, 7 appeared once.

we have to Construct set of all +ve integers 'n' such that $n \geq 5,000,000$

∴ The first digit x_1 should be 5, 6 or 7

If $x_1 = 5$ (ie) | 5 | | | | | | | then we've two 4's & one each of 3, 5, 6, 7. The no of